STRUCTURAL MODEL REPLICATION · ONE-PAGE COMPARISON

How our GPT-Sol × Qwen results differ from Manning et al.

We retained the paper's four settings—mug bargaining, bail, lawyer hiring, and an art auction—but let GPT Sol choose a new causal design and then used local Qwen agents to generate and measure the interactions.

> **The paper's experiments were more theory-anchored and behaviorally disciplined. Our frontier-planned studies were broader and more diagnostic, but the auction revealed that a stronger scientist model cannot compensate for a smaller subject model that fails to follow the game's rules.**

108 / 108	5 factors	1,073 cells	40.2 min
our interactions measured	per study in our design	across the paper's designs	our recorded model runtime

The design changed before the results did

Manning et al.

- Full factorials with variables closely tied to bargaining, judicial behavior, hiring, and auction theory.
- The auction provided a clean benchmark: prices should track the second-highest valuation.
- 1,073 treatment cells across the four published studies.

Our GPT-Sol × Qwen run

- GPT Sol chose five three-level factors and multiple outcomes for each setting.
- A 27-run orthogonal array per study estimated balanced main effects within the time budget.
- Qwen saw only its role, assigned information, and public transcript—not GPT's predictions or analysis plan.

Study-by-study differences

☕ Mug bargaining

PAPER Deal occurrence as budget, seller minimum, and attachment varied.

OURS Deal and price as budget, reservation price, attachment, outside options, and bargaining procedure varied.

Our result: higher budgets increased deals; higher seller floors reduced them. Attachment was not monotonic, while procedure and outside options mattered strongly.

⚖ Bail setting

PAPER Focused on convictions, judge workload, and defendant remorse.

OURS Used convictions, flight risk, community ties, evidence, and cooperation.

Our result: priors, flight risk, and evidence raised bail; strong ties lowered it. Cooperation produced a mixed, non-monotonic pattern.

💼 Lawyer hiring

PAPER Bar passage mattered strongly; friendliness and height were comparatively weak.

OURS Added legal reasoning, clarity, experience, and retained height as a placebo.

Our result: bar status and clarity mattered; height had no monotonic effect. Reasoning and experience unexpectedly peaked at middle levels.

🖼 Art auction

PAPER Private values yielded prices close to the second-highest valuation, matching auction theory.

OURS Added bid increments and information regimes to three bidder valuations.

Our result: only 41% of allocations were efficient; 10 cells ended with no sale, and **three winners exceeded their own assigned value**.

Bottom line

GPT Sol produced a richer experimental agenda than Qwen designed on its own: more factors, multiple outcomes, explicit placebo tests, and stronger separation between planner and subjects. But Manning et al.'s more structured experiments—especially the auction—generated cleaner theoretical behavior. The main lesson is that scientific-planning capability and simulated-subject reliability are separate bottlenecks.

Interpret cautiously

- One Qwen interaction per condition.
- Not evidence about human behavior.
- Designs differ, so this is not a controlled model comparison.
- GPT Sol was not blinded to prior project context.

Sources: Manning, Zhu & Horton, *Automated Social Science* (arXiv:2404.11794); project reports and preserved run artifacts in **structural-model-replication**.

GPT Sol scientist × Qwen3-8B subjects
Run completed October 15, 2026